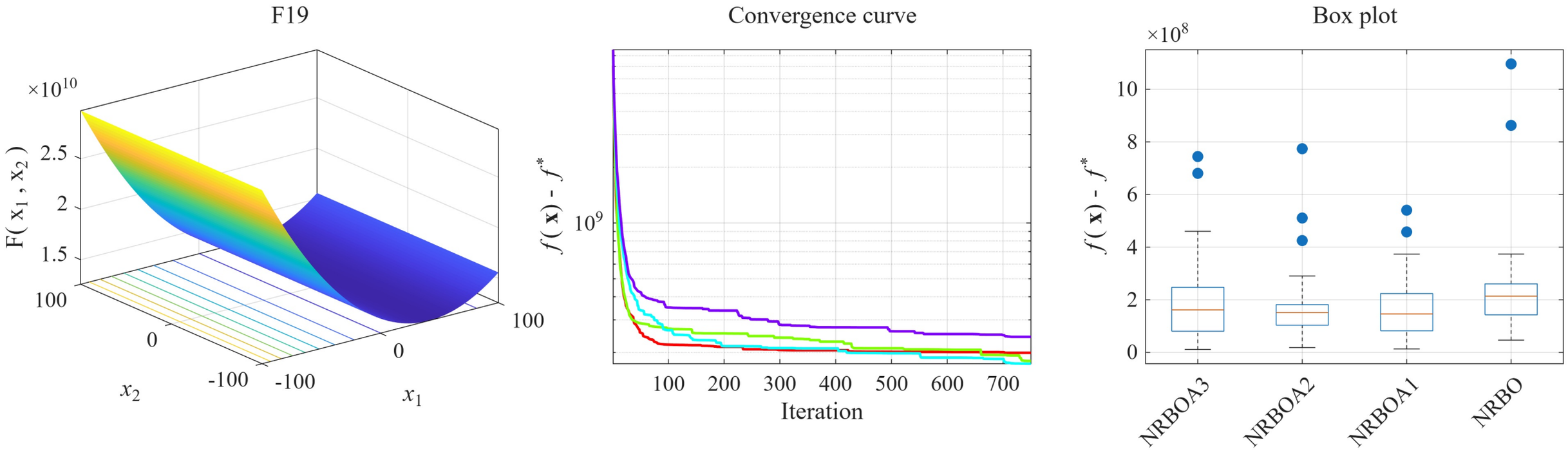

F19
×10^10
F(x1 , x2)
2.5
2
1.5
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
10^9
100
200
300
400
500
600
700
Iteration
Box plot
×10^8
10
8
6
4
2
0
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO